SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

*I **SK.ALIBABU (192524038) (**Name/Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SK.ALIBABU (192524038)

1. Constraint-Based Problem Solving using Backtracking Search

Given

Doctors = {D1, D2, D3}

Shifts = {Morning (M), Afternoon (A), Night (N)}

Constraints

1. D1 cannot work Night.
2. D2 must work before D3.
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor must be assigned exactly one shift.

Step 1: Initial State

No assignments

Doctor	Shift
D1	—
D2	—
D3	—

Step 2: Assign D1

Possible shifts:

- Morning ✔
- Afternoon ✔
- Night ✗ (Constraint 1)

Choose **Morning**.

Doctor	Shift
D1	Morning
D2	—
D3	—

Constraint Check:

- D1 not Night ✔

Step 3: Assign D2

Available shifts:

- Afternoon
- Night

Choose **Night**.

Doctor	**Shift**
D1	Morning
D2	Night
D3	—

Step 4: Assign D3

Remaining shift = Afternoon

Check constraints:

- D3 cannot work Morning
- D2 before D3

Night comes after Afternoon.

Therefore,

Night < Afternoon

Constraint violated.

Backtrack.

Step 5: Backtracking

Remove D2 = Night.

Try next value.

Assign

D2 = Afternoon

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	—

Step 6: Assign D3

Remaining shift = Night

Check constraints:

- D3 not Morning
- D2 before D3

Afternoon < Night

Only one doctor per shift

All assigned

Backtracking Tree

```
Start
│
├── D1 = Morning
│    │
│    ├── D2 = Night
│    │    │
│    │    └── D3 = Afternoon
│    │        (Backtrack)
│    │
│    └── D2 = Afternoon
│          │
│          └── D3 = Night
│
```

Final Valid Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Final Answer

Morning → D1

Afternoon → D2

Night → D3

2. Robot Navigation using BFS Search

Assume the following **5 × 5 grid**.

```
S . . X .
. X . X .
. X . . .
. . X X .
X . . . G
```

Legend:

- S = Start
- G = Goal
- X = Obstacle

Rules

- Move Up
- Move Down
- Move Left
- Move Right
- Cost = 1
- BFS used
- Heuristic = Manhattan Distance

Manhattan Distance

Formula

$h = |x1-x2| + |y1-y2|$

Goal = (5,5)

Example

For Start (1,1)

$h = |5-1| + |5-1|$

$=4+4$

$=8$

BFS Node Expansion

Queue

[S]

Expand S

Queue

[(1,2),(2,1)]

Expand (1,2)

Queue

[(2,1),(1,3)]

Expand (2,1)

Queue

[(1,3),(3,1)]

Expand (1,3)

Queue

Optimal Path

[(3,1),(2,3)]

Expand (3,1)

Queue

[(2,3),(4,1)]

Expand (2,3)

Queue

[(4,1),(3,3)]

Expand (4,1)

Queue

[(3,3),(4,2)]

Expand (3,3)

Queue

[(4,2),(3,4)]

Expand (3,4)

Queue

[(4,2),(3,5)]

Expand (3,5)

Queue

[(4,2),(4,5)]

Expand (4,5)

Queue

[(5,5)]

Goal Found.

Optimal Path

S
↓

↓

↓

→

Total Cost

↑

→

↓

↓

G

Or coordinates

(1,1)

↓

(2,1)

↓

(3,1)

↓

(4,1)

→

(4,2)

↑

(3,2)

→

(3,3)

→

(3,4)

→

(3,5)

↓

(4,5)

↓

(5,5)

Total Cost

Each move = 1

Moves = 8

Total Cost = 8

3. Autonomous Rescue Robot

(a) Constraint Analysis

Constraint 1

Robot moves only in four directions.

Effect:

- Movement is limited.
- No diagonal movement.

Constraint 2

Blocked cells cannot be crossed.

Effect:

- Robot searches for alternate routes.
- Avoids invalid paths.

Constraint 3

Limited sensing.

Effect:

- Robot knows only nearby rooms.
- Builds map while moving.

Constraint 4

Risky zones cost extra.

Safe cell = 1

Risky cell = 3

Effect:

- Robot prefers safer paths.

Constraint 5

Minimize total cost.

Effect:

- Robot balances distance and safety.

Constraint 6

Dynamic environment.

Effect:

- Robot replans whenever new information is discovered.

(b) Solution Using Uniform Cost Search (UCS)

Step 1

Start at S.

Step 2

Observe adjacent rooms.

Step 3

Insert neighboring cells into the priority queue according to cumulative cost.

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Step 4

Expand the node with the lowest total cost.

Step 5

Update costs when risky cells or obstacles are discovered.

Step 6

Repeat until Goal (G) is reached.

Example Cost

Safe move = 1

Risk move = 3

Suppose

S → Safe → Safe → Risk → Goal

Cost

1 + 1 + 3 + 1

=6

UCS chooses the least-cost path.

(c) AI Concepts

Intelligent Agent

- Perceives surroundings through sensors.
- Takes actions using actuators.
- Continuously updates knowledge.

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Rationality

A rational agent chooses the action that minimizes total cost while safely reaching the survivor.

Environment Type

Property	Description
Partially Observable	Robot cannot see the whole map
Dynamic	Environment changes during search
Sequential	Current action affects future actions
Deterministic	Each movement has a predictable result
Discrete	Rooms are represented as grid cells

(d) Justification

Uniform Cost Search (UCS) is the most suitable because:

- Finds the minimum-cost path.
- Considers additional risk costs.
- Guarantees an optimal solution when costs are non-negative.
- Works effectively in environments with varying movement costs.
- Can be combined with online updates as the robot discovers new obstacles.

Conclusion

The rescue robot uses **Uniform Cost Search with online knowledge updates** to safely navigate through the building. It intelligently avoids obstacles, minimizes travel and risk costs, dynamically updates its map, and reaches the survivor using the most cost-effective path.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandin g	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided